

Module 1: Introduction to Geometric Learning

Solutions to Exercises — Sections 1.1–1.4

Data Science Master's Programme

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These solutions cover every exercise from Sections 1.1–1.4. For each, the question is restated for convenience. Approach the solutions as a study aid: try the exercise yourself first, and only consult these to verify your reasoning or to fill gaps. Many of these exercises admit several correct lines of argument; the solutions below give one clean route, not necessarily the only one.

1 Solutions for Section 1.1 — Symmetries and Group Actions

Exercise 1.1: 1.1 — $(\mathbb{R} \setminus \{0\}, \times)$ is a group

Verify that $(\mathbb{R} \setminus \{0\}, \times)$ is a group. Identify the identity and the inverse of an arbitrary element.

Solution 1.1

Check the four axioms:

- (i) *Closure*: if $a, b \neq 0$ then $ab \neq 0$, so $ab \in \mathbb{R} \setminus \{0\}$. ✓
- (ii) *Associativity*: multiplication of reals is associative. ✓
- (iii) *Identity*: $e = 1$ satisfies $1 \cdot a = a \cdot 1 = a$ for all a . ✓
- (iv) *Inverse*: for $a \neq 0$ the inverse is $a^{-1} = 1/a \in \mathbb{R} \setminus \{0\}$ and $a \cdot (1/a) = 1$. ✓

Note: we excluded 0 precisely because 0 has no multiplicative inverse.

Exercise 1.2: 1.1 — Action of $\{\pm 1\}$ on $\mathbb{R}^2$

Let $G = \{1, -1\}$ under multiplication and $X = \mathbb{R}^2$, with action $g \cdot (x, y) = (gx, gy)$. Describe each group element geometrically, and draw the orbits of $(1, 0)$ and $(0, 0)$.

Solution 1.2

- $1 \cdot (x, y) = (x, y)$ — the identity map (no movement).
- $-1 \cdot (x, y) = (-x, -y)$ — point reflection through the origin (equivalent to a 180 rotation).

Orbits.

- Orbit of $(1, 0)$: $\{(1, 0), (-1, 0)\}$ — a pair of antipodal points on the x -axis.
- Orbit of $(0, 0)$: $\{(0, 0)\}$ — a single point (the origin is a *fixed point* of the entire group, with the full group as stabiliser).

More generally the orbit of any $(a, b) \neq (0, 0)$ has size 2; the only orbit of size 1 is the origin.

Exercise 1.3: 1.1 — Symmetries of molecular graphs

Why must a molecular property predictor be S_n -invariant but *not* necessarily $\text{SO}(3)$ -invariant if 3D coordinates are absent?

Solution 1.3

S_n -invariance is required. A graph encodes the molecule via an adjacency matrix A together with node features. The choice of how to enumerate atoms is arbitrary — atom 1 might be the leftmost or the rightmost carbon, but the molecule (and hence its physical properties: solubility, toxicity, dipole moment) does not depend on this labelling. Any permutation $\sigma \in S_n$ acts as $A \mapsto P_\sigma A P_\sigma^\top$, and the predictor's output must be invariant under this action.

$\text{SO}(3)$ -invariance is irrelevant without 3D coordinates. $\text{SO}(3)$ acts by rotating 3D atomic positions. If the model receives *only* the graph (no Cartesian coordinates), there is no geometric variable on which $\text{SO}(3)$ could act in the first place — the question of $\text{SO}(3)$ -invariance is vacuous.

If 3D coordinates *were* provided, $\text{SO}(3)$ -invariance would become necessary (a molecule rotated in space is the same molecule).

Exercise 1.4: 1.1 — Order of the dihedral group

Compute $|D_n|$ (symmetries of a regular n -gon) and decide for which n it is abelian.

Solution 1.4

The symmetries of a regular n -gon consist of:

- n rotations by $2\pi k/n$ for $k = 0, 1, \dots, n-1$;
- n reflections across axes through the centre.

Hence $|D_n| = 2n$.

Abelian-ness. For $n = 1$ and $n = 2$: $D_1 = \mathbb{Z}/2$ and $D_2 = \mathbb{Z}/2 \times \mathbb{Z}/2$ (the Klein four-group), both abelian.

For $n \geq 3$: let r be the rotation by $2\pi/n$ and s a reflection. Then $sr s = r^{-1}$, i.e. $sr \neq rs$ (unless $r^2 = e$, which is the $n \leq 2$ case). So D_n is non-abelian for $n \geq 3$.

Exercise 1.5: 1.1 — Orbit size via the orbit–stabiliser theorem

C_4 acts on a digit image with trivial stabiliser. Find the orbit size.

Solution 1.5

The *orbit–stabiliser theorem* states

$$|G \cdot x| = \frac{|G|}{|\text{Stab}(x)|}.$$

Here $G = C_4$ has order 4, and the stabiliser is trivial ($|\text{Stab}(x)| = 1$), so the orbit has size $|G \cdot x| = 4$.

Interpretation. The four images $\{x, R_{90}x, R_{180}x, R_{270}x\}$ are all distinct — none of them is equal to the original under a non-trivial rotation. A digit with trivial C_4 -stabiliser is one that has *no* four-fold rotational symmetry; this is generic for handwritten digits, with the partial exception of the digit “8”, which is approximately C_2 -symmetric (not C_4).

Exercise 1.6: 1.1 — Challenge: $\text{SO}(2)$ is a group

Show that $\{Q \in \mathbb{R}^{2 \times 2} : Q^\top Q = I, \det Q = +1\}$ forms a group under matrix multiplication.

Solution 1.6

Denote the set by $\text{SO}(2)$. We verify the four axioms.

Closure. If $Q_1, Q_2 \in \text{SO}(2)$ then $(Q_1 Q_2)^\top (Q_1 Q_2) = Q_2^\top Q_1^\top Q_1 Q_2 = Q_2^\top I Q_2 = I$, and $\det(Q_1 Q_2) = \det(Q_1) \det(Q_2) = 1 \cdot 1 = 1$. So $Q_1 Q_2 \in \text{SO}(2)$.

Associativity. Inherited from matrix multiplication.

Identity. I satisfies $I^\top I = I$ and $\det I = 1$, so $I \in \text{SO}(2)$ and it acts as the identity.

Inverses. For $Q \in \text{SO}(2)$, $Q^\top Q = I$ means $Q^{-1} = Q^\top$. Check $Q^\top$ is in $\text{SO}(2)$: $(Q^\top)^\top Q^\top = Q Q^\top = I$ (since $Q^\top = Q^{-1}$, both products equal I), and $\det Q^\top = \det Q = 1$. So $\text{SO}(2)$ is a group. (Geometrically, its elements are the rotations R_θ of $\mathbb{R}^2$ about the origin.)

2 Solutions for Section 1.2 — Invariance and Equivariance

Exercise 2.1: 1.2 — Sign action on $\mathbb{R}$

$G = \{+1, -1\}$ acts on $\mathbb{R}$ by multiplication. Classify (a) x^2 , (b) $|x|$, (c) x^3 , (d) e^x as invariant, equivariant, or neither.

Solution 2.1

Recall the action: $g \cdot x = gx$. For each function we check whether $f(gx) = f(x)$ (invariant) or $f(gx) = g \cdot f(x) = gf(x)$ (equivariant with the same action on the codomain).

- (a) $f(x) = x^2$: $f(-x) = (-x)^2 = x^2 = f(x)$. **Invariant.** (Also vacuously equivariant.)
- (b) $f(x) = |x|$: $f(-x) = |-x| = |x| = f(x)$. **Invariant.**
- (c) $f(x) = x^3$: $f(-x) = -x^3 = (-1) \cdot f(x)$. **Equivariant** but not invariant — it is an *odd* function.
- (d) $f(x) = e^x$: $f(-x) = e^{-x}$, which is in general neither e^x nor $-e^x$. **Neither** invariant nor equivariant.

Exercise 2.2: 1.2 — Set MLP

$f(X) = \sigma(W \cdot \text{mean}(X) + b)$. Invariant or equivariant under permutations?

Solution 2.2

The mean $\frac{1}{n} \sum_i x_i$ is itself a permutation-invariant quantity of X : reordering the x_i does not change the sum. Therefore $W \cdot \text{mean}(X) + b$ is unchanged, and so is $\sigma(\cdot)$.

Conclusion: f is S_n -**invariant**.

Remark. This is the basic DeepSets template (Zaheer et al., 2017): any function of the form $\rho(\sum_i \phi(x_i))$ is permutation-invariant, and conversely all continuous permutation-invariant functions on sets of bounded cardinality can be written in this form (DeepSets universal approximation theorem).

Exercise 2.3: 1.2 — Conv layer is translation- but not rotation-equivariant

Why is a 3×3 convolution translation-equivariant but not rotation-equivariant?

Solution 2.3

Translation equivariance. The convolution operation is

$$(f * x)(i, j) = \sum_{u, v} f(u, v) x(i + u, j + v).$$

If we replace x by its translate $x_\delta(i, j) = x(i - \delta_1, j - \delta_2)$, a change of indices in the sum shows $(f * x_\delta)(i, j) = (f * x)(i - \delta_1, j - \delta_2)$ — exactly a translation of the feature map by δ . Hence equivariance.

Rotation non-equivariance. A 3×3 filter has 9 scalar weights with no symmetry imposed: $w_{11}, w_{12}, \dots, w_{33}$. Rotating the input image by 90 permutes the locations of the input pixels relative to the filter, but the filter weights themselves do not rotate: the filter pattern looks at fundamentally different spatial relationships. Consequently the rotated output is *not* a rotation of the unrotated output.

Quick demonstration: take a filter $f = [[1, 0, 0], [0, 0, 0], [0, 0, 0]]$ (extract top-left neighbour). Applied to an input, it produces one feature map. Rotating the input by 90 and applying the same filter extracts a different pixel relation entirely — the result is not a rotation of the original feature map.

Architectural fix. *Group equivariant CNNs* (G-CNNs, Cohen and Welling, 2016) achieve rotation equivariance by tying the weights of the filter across a discrete group of rotations and considering features as functions on $\mathbb{Z}^2 \times C_n$ rather than on $\mathbb{Z}^2$ alone.

Exercise 2.4: 1.2 — Continuous $\text{SO}(2)$ -invariant functions on $\mathbb{R}^2$

Show that any continuous $\text{SO}(2)$ -invariant $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ depends only on $\|x\|$.

Solution 2.4

Strategy. An $\text{SO}(2)$ -invariant function is constant on each orbit of $\text{SO}(2)$, and the orbits of $\text{SO}(2)$ on $\mathbb{R}^2$ are exactly the concentric circles centred at the origin (plus the origin itself).

Proof. Let $r \geq 0$. The set of points at distance r from the origin is a single $\text{SO}(2)$ -orbit:

$$O_r = \{x \in \mathbb{R}^2 : \|x\| = r\}.$$

By $\text{SO}(2)$ -invariance, f is constant on O_r . Define

$$h(r) := f(r, 0).$$

Then for any $x = (x_1, x_2)$ with $\|x\| = r$, there exists $R_\theta \in \text{SO}(2)$ with $R_\theta(r, 0) = x$, and

$$f(x) = f(R_\theta(r, 0)) = f(r, 0) = h(r) = h(\|x\|).$$

Continuity of h follows from continuity of f restricted to the positive x -axis $\{(r, 0) : r \geq 0\}$. This is the simplest example of the principle “invariant functions = functions on the orbit space”. Here the orbit space $\mathbb{R}^2/\text{SO}(2) \cong \mathbb{R}_{\geq 0}$.

Exercise 2.5: 1.2 — Challenge: linear equivariant maps for $\mathbb{Z}/2$

$G = \mathbb{Z}/2\mathbb{Z}$ acts on $\mathbb{R}^2$ by $1 \cdot v = -v$. Find all linear $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that are equivariant.

Solution 2.5

Let $f(v) = Av$ for some matrix $A \in \mathbb{R}^{2 \times 2}$. Equivariance under $1 \cdot v = -v$ requires

$$A(-v) = -A(v) \quad \text{for all } v \in \mathbb{R}^2.$$

But $A(-v) = -Av$ holds for *any* linear map (linearity gives $A(-v) = -Av$ automatically). Therefore **every linear map** $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is equivariant under this action.

Interpretation. The action of $\mathbb{Z}/2$ here is the same representation on both source and target (the sign representation). By Schur’s Lemma, the equivariant maps form the commutant of this representation — and for a one-dimensional representation appearing twice in the decomposition of $\mathbb{R}^2$, the commutant is the full 2×2 matrix algebra. (Both copies of $\mathbb{R}$ transform under the same sign character, so any linear combination of them is also $\mathbb{Z}/2$ -equivariant.)

Compare: if the codomain action had been trivial ($g \cdot w = w$), then equivariance would require $A(-v) = Av$, i.e. $Av = -Av$ for all v , forcing $A = 0$.

3 Solutions for Section 1.3 — Manifold Learning

Exercise 3.1: 1.3 — Euclidean distance fails on the Swiss roll

Why does Euclidean distance fail on a Swiss roll? Sketch two geodesically distant but Euclidean-close points.

Solution 3.1

The Swiss roll is a 2D sheet rolled up in 3D. As the sheet wraps around, two layers of the sheet that are far apart *along the manifold* can become physically close in $\mathbb{R}^3$ because they sit on top of each other.

Example points. Place one point near the inner part of the roll, on its outer surface, and a second point on the next outer layer of the roll, directly above it. Their straight-line distance in $\mathbb{R}^3$ is small (the layers are thin), but to travel from one to the other *along the surface* you must traverse most of the roll's length.

Consequence. Any algorithm that uses Euclidean distance (clustering, k -NN, classical MDS) will mistakenly treat these points as similar, and downstream tasks (classification, regression, density estimation) will be confused.

Exercise 3.2: 1.3 — Choice of k in Isomap

What can go wrong if k is too large? Too small?

Solution 3.2

Too large. *Short-circuit edges.* If k is large enough that the neighbourhood graph connects points across different manifold layers (e.g. across two layers of the Swiss roll), the shortest path between two points may follow the short-circuit edge rather than the true geodesic on the manifold. Estimated geodesic distances become drastically wrong and the embedding collapses layers together.

Too small. *Disconnected graph.* If k is so small that the neighbourhood graph fragments into multiple connected components, shortest paths between components are undefined (infinite), MDS cannot proceed, and Isomap fails or returns an embedding only of the largest connected component.

Practical rule. Cross-validate k on a downstream task or use the smallest k for which the graph remains connected. Typical values: $k = 5$ to 20 , depending on dataset density.

Exercise 3.3: 1.3 — LLE constraint $\sum_j w_{ij} = 1$

Why does LLE impose $\sum_j w_{ij} = 1$? What invariance does it enforce?

Solution 3.3

The constraint $\sum_j w_{ij} = 1$ makes the reconstruction weights invariant to *affine transformations* of the local neighbourhood — specifically, to translations.

Translation invariance. If we shift every point by a constant $t \in \mathbb{R}^d$, the reconstruction

equation becomes

$$x_i + t = \sum_j w_{ij}(x_j + t) = \sum_j w_{ij}x_j + t \sum_j w_{ij}.$$

The right-hand side equals $x_i + t$ precisely when $\sum_j w_{ij} = 1$. So the same weights w_{ij} are reconstruction weights both before and after translation.

Rotation and scaling invariance follow automatically: rotation and scaling are linear maps that commute with the linear combination $\sum_j w_{ij}x_j$.

Consequence. The local structure captured by the weights is intrinsic to the manifold's local geometry — independent of the arbitrary global coordinate frame in $\mathbb{R}^D$. This is exactly the property LLE needs in order for the embedding step to use the same weights in the low-dimensional space.

Exercise 3.4: 1.3 — Circle dataset

1000 noisy points on $S^1 \subset \mathbb{R}^2$. Which of Isomap, LLE, Diffusion Maps recovers the circular structure most robustly?

Solution 3.4

All three can recover the 1D structure, but their robustness differs.

Isomap. The circle is simply-connected (in the sense of having no holes *in the dataset*; topologically S^1 has $\pi_1 = \mathbb{Z}$). Isomap assumes a simply-connected manifold and struggles with the circular topology: geodesic distance is well defined but classical MDS embeds the circle in $\mathbb{R}^2$, not $\mathbb{R}^1$, so a 2D embedding will recover a circular arrangement but a strict 1D embedding will not.

LLE. Locally linear reconstruction handles the circle's topology well — LLE only needs local linearity, which S^1 has everywhere. With Gaussian noise the choice of k matters: too small and noise dominates, too large and weights wrap around the circle.

Diffusion Maps. Random walk on the noisy circle is the most robust. The Markov matrix P 's top non-trivial eigenvectors are approximately cos and sin functions of the position along the circle, giving an exact 2D embedding that is a (slightly distorted) circle. Diffusion time $t > 1$ averages out Gaussian noise.

Recommendation. Diffusion Maps are the most robust on a noisy circle — they natively handle non-trivial topology and have built-in denoising via the time parameter t .

Exercise 3.5: 1.3 — Misreading t -SNE distances

Why are distances between t -SNE clusters not meaningful? Sketch a case where t -SNE and Diffusion Maps disagree.

Solution 3.5

Why distances are not meaningful. The t -SNE loss is $\text{KL}(P\|Q)$, which is dominated by terms where p_{ij} is large — i.e. by *neighbouring* point pairs in the high-dimensional space. Once two points are not neighbours in high dimensions, p_{ij} is essentially zero and the KL divergence places almost no penalty on their separation in the embedding. As a result, the positions of well-separated clusters are determined almost entirely by

the optimisation dynamics (initial conditions, learning rate, repulsive force interactions between clusters) rather than by any property of the data.

Furthermore, the heavy-tailed Student- t kernel actively pushes dissimilar points apart, equalising cluster separations regardless of their true high-dimensional distances.

Disagreement example. Consider data forming a 1D curve in high-dimensional space, with three Gaussian density bumps along the curve (slight clustering, but the points are connected along the curve).

- *Diffusion Maps* embed the curve as a continuous arc, with the three density bumps appearing as denser regions along it. The curve's topology and ordering are preserved.
- *t-SNE* typically breaks the curve into three disconnected blobs (because moderate-distance neighbours along the curve get pushed apart by the heavy-tailed repulsion), creating the visual illusion of three discrete clusters and obscuring the underlying 1D continuum.

This kind of disagreement is well documented in trajectory inference in single-cell genomics, where *t-SNE* can fragment a developmental continuum into spurious clusters while Diffusion Maps reveal it as a connected branching structure.

Exercise 3.6: 1.3 — Perplexity sweep on MNIST

Describe how a *t-SNE* plot of MNIST changes as perplexity sweeps from 5 to 200. Why does very low perplexity create the illusion of many small clusters?

Solution 3.6

Low perplexity (5). Each point sees only its 4–5 closest neighbours. *t-SNE* preserves micro-neighbourhoods aggressively but has no information about medium-scale structure. The embedding fragments into many small isolated blobs corresponding to local density peaks — including spurious clusters that are just small groups of similar digits within the same true class.

Moderate perplexity (30, the default). Each point sees about 30 neighbours — typically including most variations of the same digit class. The embedding shows 10 well-separated clusters approximately corresponding to the 10 digits.

High perplexity (100–200). The notion of “neighbour” expands toward global structure. Clusters bleed into each other. At very high perplexity ($n/2$ or above) *t-SNE* essentially reduces to MDS on the full distance matrix and the cluster structure is lost.

Why low perplexity hallucinates clusters. With perplexity ~ 5 , each point's Gaussian bandwidth σ_i becomes very small. Points that lie even slightly outside the 5-neighbourhood receive near-zero p_{ij} , and the Student- t repulsion pushes them away aggressively. The result: any local dip in density appears as a cluster boundary, even when no real separation exists in the data. This is the central caveat of the Distill.pub article on *t-SNE*.

Practical advice. Always run *t-SNE* at multiple perplexities (e.g. 5, 30, 100) and only trust structure that persists across all three.

Exercise 3.7: 1.3 — t -SNE vs. UMAP at scale

Compare t -SNE and UMAP via the attract/repel trade-off. For embedding 10^6 points into $\mathbb{R}^{10}$ for downstream classification, which method and why?

Solution 3.7

UMAP — for several converging reasons.

- (i) **Scalability.** UMAP uses approximate k -NN (NN-Descent) and *negative sampling* for the repulsive term, making it roughly 5–10 $\times$ faster than Barnes–Hut t -SNE and able to handle 10^6+ points routinely.
- (ii) **Target dimension.** t -SNE is effectively restricted to 2D/3D — its Student- t kernel was hand-designed for low-dimensional visualisation. UMAP can embed into any dimension; $\mathbb{R}^{10}$ is fine.
- (iii) **Global structure for downstream learning.** A classifier operating on the embedded features benefits from preserved global geometry: similar classes should appear close to each other in feature space. UMAP’s cross-entropy objective preserves more global structure than t -SNE’s KL.
- (iv) **Reproducibility and feature stability.** UMAP is more reproducible across random seeds than t -SNE, which is important if the embedding is used as a fixed feature transform.

Modern alternatives to consider. For very large datasets where embedding quality matters, contrastive self-supervised methods (SimCLR, DINO) often outperform UMAP because their attractive force is defined by data augmentation rather than by k -NN, giving a learned (rather than algorithmic) notion of similarity. The framework of contrastive neighbour embedding ties all these methods together.

Exercise 3.8: 1.3 — Challenge: trivial eigenvector of P

Why is the all-ones eigenvector of P discarded in Diffusion Maps? What does it represent?

Solution 3.8

The eigenpair. Since P is row-stochastic ($\sum_j P_{ij} = 1$ for all i), the vector $\mathbf{1} = (1, 1, \dots, 1)^\top$ satisfies $P\mathbf{1} = \mathbf{1}$, so it is a right eigenvector with eigenvalue $\lambda_0 = 1$.

Why discard it. The diffusion map embeds point x_i as $\Psi_t(x_i) = (\lambda_1^t \psi_1(i), \lambda_2^t \psi_2(i), \dots)$, where the eigenvalues are sorted in decreasing order. If we included the $\lambda_0 = 1$ eigenvector $\psi_0 = \mathbf{1}$, every point x_i would receive the same value $\psi_0(i) = 1$ in that coordinate. This coordinate carries *no information about position on the manifold* — it just adds a constant to every embedded point.

Geometric meaning. The all-ones eigenvector corresponds to the *stationary distribution* of the random walk: in the limit $t \rightarrow \infty$, the walker forgets its starting point and the distribution over states converges to the stationary one. The trivial eigenvector encodes this “forgetting” — the equilibrium configuration that contains no geometric information beyond the fact that the chain is ergodic.

Analogy. This is exactly analogous to discarding the constant function $\phi = 1$ when

computing the Fourier series of a signal: the constant Fourier coefficient is the average, which carries no oscillatory information.

4 Solutions for Section 1.4 — Gauge Theories

Exercise 4.1: 1.4 — $\text{SO}(2)$ is insufficient for S^2

Why is a global rotation group $\text{SO}(2)$ insufficient on S^2 ? What goes wrong at the poles?

Solution 4.1

$\text{SO}(2)$ acts by a single rotation angle. To define a rotation, you must choose a reference direction — and on a curved surface, no such global direction exists.

Concretely. The longitude–latitude frame defines “East” and “North” at every point of S^2 , but at the North Pole every outward direction is “South” and the notion of “East” is ill-defined (all longitude lines meet there). The same coordinate system that is locally smooth becomes *singular* at the poles.

Topological reason. The *hairy ball theorem* states that no continuous non-vanishing tangent vector field exists on S^2 . Hence no globally consistent choice of “one direction at every point” is possible. A single global rotation cannot describe the symmetries of structures (such as wind fields) defined on S^2 because there is no global frame for it to act on.

Resolution. Use a *local* rotation group acting independently at each point — i.e. a gauge group $C^\infty(\mathcal{M}, \text{SO}(2))$. This is the central idea of Section 1.4.

Exercise 4.2: 1.4 — Scalar and vector fields on S^2

Is temperature on Earth a section of TS^2 or of the trivial bundle $S^2 \times \mathbb{R}$? What about wind velocity?

Solution 4.2

Temperature: trivial scalar bundle. Temperature $T(p) \in \mathbb{R}$ is a scalar — a single real number attached to each point on S^2 . It does not change under a change of local frame (you do not need to know which way East is to read a thermometer). It is therefore a section of the trivial bundle $S^2 \times \mathbb{R}$.

Wind velocity: tangent bundle. Wind velocity $\mathbf{v}(p)$ is a tangent vector at p : it has a direction *along the surface*. Its representation in components (v_E, v_N) depends on the local frame; under a change of frame at p , the components transform by an $\text{SO}(2)$ matrix. This is the signature of a section of TS^2 .

Consequence. On S^2 , any non-vanishing wind field must have at least one zero (eye of a storm at the North Pole, etc.) by the hairy ball theorem — a tangible meteorological consequence of the topology of the sphere.

Exercise 4.3: 1.4 — Message passing without parallel transport

What happens if a mesh message-passing network ignores the connection? Where would the error be worst?

Solution 4.3

What happens. Each node has a local frame (a tangent basis), and feature vectors are encoded in components with respect to that frame. “Aggregating features” by averaging the component vectors of neighbours — without first transporting them to the central node’s frame — mixes components expressed in incompatible coordinate systems. The resulting feature is geometrically meaningless: its components have no consistent interpretation, and equivariance is destroyed.

Empirically (cf. the notebook for Section 1.4), naïve aggregation produces a vector that:

- is not tangent to the manifold (points partly into the ambient embedding space);
- has a magnitude that depends on the arbitrary frame choices, breaking gauge invariance;
- changes value if the network is re-run with rotated local frames.

Where the error is worst.

1. *Highly curved regions.* Where adjacent frames differ by a large rotation, ignoring the connection causes large mis-aggregation.
2. *Coarse meshes.* Larger triangles mean larger inter-frame holonomies between neighbours.
3. *Many message-passing layers.* Errors compound across layers — a deep network without frame correction will accumulate substantial drift.

For nearly-flat regions of the mesh with very fine triangulation, the difference between adjacent frames is small and the naïve aggregation is approximately correct.

Exercise 4.4: 1.4 — Hairy ball does not apply to T^2

Sketch why the hairy ball theorem fails on the torus $T^2 = S^1 \times S^1$. What does this say about frames on T^2 ?

Solution 4.4

Why the theorem fails. The hairy ball theorem states that any continuous tangent vector field on S^2 must vanish somewhere. The proof rests on the fact that the *Euler characteristic* of S^2 is $\chi(S^2) = 2 \neq 0$, and the Poincaré–Hopf theorem ties the sum of indices of zeros of a vector field to χ . So any vector field on S^2 has total index 2, forcing the existence of zeros.

For the torus, $\chi(T^2) = 0$. Therefore there is no obstruction to the existence of a continuous non-vanishing tangent vector field on T^2 . Indeed, the field $\partial/\partial\theta_1$ (“always go around the first S^1 at unit speed”) is a continuous, non-vanishing vector field that lives globally on the whole torus.

Consequence for frames. The torus is *parallelisable*: there exists a global frame $\{e_1, e_2\}$

for TT^2 — namely $(\partial/\partial\theta_1, \partial/\partial\theta_2)$. So gauge theory on the torus is *globally trivialisable*: we can pick one global frame and the connection on TT^2 (with respect to the flat metric) has zero holonomy around contractible loops.

This makes the torus a degenerate case where the full gauge machinery is not strictly necessary — flat tori are the simplest manifolds on which to test gauge-equivariant code, since results should agree with naïve calculations.

Exercise 4.5: 1.4 — When do two paths induce the same transport?

When do two paths γ_1, γ_2 from p to q induce the same parallel transport?

Solution 4.5

Concatenation argument. Consider the closed loop $\gamma = \gamma_2^{-1} \circ \gamma_1$: go from p to q via γ_1 , then back from q to p via γ_2 reversed. This loop is based at p .

Parallel transport is a group homomorphism in the path concatenation structure:

$$\Pi_{\gamma_2^{-1} \circ \gamma_1} = \Pi_{\gamma_2}^{-1} \circ \Pi_{\gamma_1}.$$

So

$$\Pi_{\gamma_1} = \Pi_{\gamma_2} \iff \Pi_{\gamma_2^{-1} \circ \gamma_1} = \text{id}_{F_p}.$$

That is: the two paths give the same transport *iff the composite loop has trivial holonomy*.

Sufficient condition: flat connection on a simply-connected region. If the connection is flat (curvature $F = 0$) and the loop lies in a simply-connected region, the holonomy is trivial. So on a flat manifold (e.g. $\mathbb{R}^d$ with the standard connection), any two paths give the same transport — which is why we never worry about this on Euclidean space.

General principle. The holonomy is path-independent only modulo the *holonomy group* of the connection, which is a Lie subgroup of the structure group measuring the total “twisting” picked up by closed loops.

Exercise 4.6: 1.4 — Challenge: Möbius bundle and gauge triviality

Can the Möbius bundle over S^1 be made trivial by a gauge transformation $g(\theta) \in \{+1, -1\}$?

Solution 4.6

No.

Argument via global sections. A trivial bundle $S^1 \times \mathbb{R}$ admits a continuous non-vanishing global section, e.g. $\theta \mapsto (\theta, 1)$.

The Möbius bundle, by contrast, does *not* admit a continuous non-vanishing global section. Heuristically: any continuous section $s : S^1 \rightarrow E_{\text{Möb}}$ must satisfy $s(\theta + 2\pi) = -s(\theta)$ because of the twist. If s is continuous on $[0, 2\pi]$ and non-vanishing, intermediate value theorem applied to (a continuous branch of) s shows that $s(\theta + 2\pi) = -s(\theta)$ forces s to vanish somewhere.

More precisely: orient $\mathbb{R}$ by sign. A non-vanishing section takes values in either $\mathbb{R}_{>0}$ or $\mathbb{R}_{<0}$ in any trivialising chart. Going around S^1 once, the twist forces $s(2\pi)$ to be the negative of $s(0)$. So if $s(0) > 0$ then $s(2\pi) < 0$, and by continuity s must hit 0 somewhere.

— a contradiction.

Why a gauge transformation cannot fix this. A gauge transformation $g : S^1 \rightarrow \{+1, -1\}$ acts by sign flipping the fiber at each point. A continuous map from S^1 (connected) to the discrete two-point set $\{+1, -1\}$ must be constant. So $g \equiv +1$ or $g \equiv -1$, neither of which changes the topology of the bundle.

Equivalently: bundles up to gauge equivalence are classified by topological invariants (here the first Stiefel–Whitney class $w_1 \in H^1(S^1; \mathbb{Z}/2) \cong \mathbb{Z}/2$). The trivial bundle has $w_1 = 0$, the Möbius bundle has $w_1 = 1$, and gauge transformations preserve this invariant. They are genuinely different bundles.

Closing notes

These solutions are intended as a baseline. Many exercises (especially the proof-flavoured ones in Section 1.2 and 1.4) admit multiple correct strategies — for instance, the orbit-stabiliser theorem can be proved via cosets or via a direct counting argument; Schur’s Lemma underlies several proofs that we instead handled by direct computation. Where this is the case, the solution above gives the shortest clean route, but you should feel free to explore alternatives.

For Section 1.3 in particular, the exercises on t -SNE and UMAP benefit enormously from *running* the methods on real data. The interactive Distill.pub article on t -SNE and the UMAP documentation’s “How UMAP Works” page are required complements to the solutions above.